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Polymer Foams Made of Immiscible Polymer Blends Compatibilized by Janus Particles—Effect of Compatibilization on Foam Morphology**

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Polymer blends have the potential to be homogeneously foamed, since the interface area between their phases can be used as bubble nucleating agents during the foaming process. Immiscible polymer blends of poly(2,6-dimethyl-1,4-phenylene ether) PPE and poly(styrene-co-acrylonitrile) SAN with a weight ratio of 60/40 are compatibilized by different amounts (1–10 wt%) of Janus nanoparticles (JPs). Here, we report on the nucleation effect of JPs and the dispersed PPE phase on the foaming behavior of PPE/SAN blends. The plasticization of CO₂ and its solubility and diffusion behavior in the PPE/SAN blend are analyzed and correlated to blend morphologies. It is shown that partial coverage of the raspberry structure results in stress concentration at the blend interface during the saturation step. This formation of new surfaces by cleavage (and maybe microcrazes) results in a higher sorption of the blowing agent, which affects the foaming behavior. Through temperature-induced batch foaming with CO₂, the blends are foamed to find the optimum foaming conditions and relationships between the morphology of the compact and foamed blends (cellular structure) and the density are established. Finally, it was concluded that by using Janus particles, homogenous foams with small cell sizes, and relatively high densities are produced.

1. Introduction

Blending is one of the methods to produce polymer materials with tailored properties chosen from two or more of

its polymer components. However, most polymers are immiscible, and tend to form macro-phase separated regions after blending, yet a decreased homogeneity of the final blend negatively affects its properties. Using small amounts of materials known as compatibilizers, will help to preserve the desired properties and in some cases promotes synergistic effects, e.g., compatibilization in combination with enhancing the toughness of the system. The immiscible polymer blend of poly(2,6-dimethyl-1,4-phenylene ether), PPE, and poly(styrene-co-acrylonitrile), SAN, promises a good combination with advantages like high heat distortion temperature, toughness, good flame resistance, and low-moisture absorption of PPE with good properties of SAN, such as chemical resistance, good processability, and low cost.^[1–4] The application of PPE is limited by its low oxidation resistance and high viscosity, which aggravates the processing. Therefore, the polymer is often blended with SAN, polystyrene, high-impact polystyrene, polyamide, or polyethylene.^[4] However, SAN containing 19 wt% acrylonitrile is immiscible with PPE, causing internal stresses and inhomogeneity in the matrix, and makes an additional compatibilizer necessary to improve stress transfer, decrease particle size of the dispersed phase, and stabilize the morphology during processing.

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Foaming polymer blends has a high potential for a better control over properties such as the size of the cellular structure, open or closed cell content and properties, as well as foam density.^[5–7] Due to those properties, foams can be used in applications such as energy absorption, thermal and acoustic insulation, and packaging. Foaming polymer blends can be very challenging due to their complex nature and several parameters influencing the foaming behavior of the involved compounds. Especially in terms of immiscible polymer blend systems, the foaming process is greatly influenced by the morphology and the interactions between the phases. The influence of different linear polystyrene-*block*-polybutadiene-*block*-poly(methyl methacrylate) (PS-*b*-PB-*b*-PMMA, SBM) triblock terpolymers on the foaming behavior of PPE/SAN blends using different blowing agents has been previously investigated.^[8–11] They could either achieve low foam density of about 200 kg m^{-3} with large cell sizes using organic solvents as blowing agents, or nanocellular foams with large densities of 600 kg m^{-3} using CO_2 as blowing agents. The heterogeneous nucleation, which is favored to produce a more homogenous, finer foam structure, can occur at the interphase between the two blend phases.^[7,12] During nucleation and foaming process, the dissolution of the blowing agent in the highly viscous PPE does not lead to foaming, but most likely interfacial enrichment of CO_2 between the PPE and SAN.^[9] Also, differences in glass transition temperature of amorphous polymers can be used in the batch process for selective foaming of one phase, which was studied more intensely by Ohshima and co-workers^[13–15] They reached nanocellular foam structures from PS/PMMA blends produced via the pressure quench method with CO_2 and achieved a fine blend structure of 40–50 nm, where the cell nucleation was localized exclusively in the PMMA domains.^[15] In another work, cell diameters of 0.5–2 μm were achieved in a polypropylene (PP)/propylene-ethylene copolymer (PER) blend system by adjusting the blend ratio, depressurization rate, and foaming temperature.^[13] Gong and Ohshima investigated the effect of interfacial tension on the bubble nucleation in PMMA/polycarbonate (PC) blends and found out that by reduction of the interfacial tension, a better distribution of the dispersed phase and an enhanced bubble nucleation is reached.^[16] The compatibilizer in the immiscible blend system reduces the stress at the interface, affecting the foam morphology, the interfacial tension, and the foaming behavior of these materials.^[17–20]

While blending, one should notice that the morphology of the blend is greatly influenced by the blend composition, viscosity of the blends, compatibilizer type and amount, processing conditions, and many other factors.^[21–23] In the last few decades, the effect of compatibilization on the morphology of immiscible blends has been thoroughly studied.^[1,19,22–26] Wu^[23] was one of the first to investigate the formation of dispersed phase structures in immiscible polymer blends and determined that the droplet size is directly proportional to the interfacial tension and therefore, decreases if the interfacial tension decreases. Compatibilizers are materials that have a

high interfacial activity and hence lower the interfacial tension between the blend components and promote homogeneity. These materials can greatly influence the blend morphology. For example, PPE/SAN blends compatibilized with SBM triblock terpolymers show an interesting 3D morphology (also known as raspberry morphology) reported for the first time in 1993 by Auschra and Stadler.^[1,27]

In the raspberry morphology, the SBM triblock terpolymers are located at the interface between the PPE and SAN blend phases, as the PS end block is miscible with PPE and the PMMA end block is miscible with SAN.^[1] Due to the immiscibility of the PB middle block with both PPE and SAN phases, it appears exactly at the interface while having strong bonding to both phases. This 3D structure looks like raspberry-shaped islands of the PB covered dispersed phase in the matrix.^[1] The effect of SBM compatibilization on the morphology during the extrusion process, as well as the influence of the compatibilizer block length on the morphological and mechanical properties has been thoroughly investigated.^[28–30] Recently, a novel class of particulate compatibilizers based on SBM triblock terpolymers, known as Janus particles,^[31] have been developed. These particles are produced from SBM triblock terpolymers that are arranged in multi-compartment micelles and, by developing a new facile production method,^[32] are available in larger amounts. It has been shown that due to their special structure, they show higher surface activity and lower interfacial tensions compared to the SBM triblock terpolymers when used as stabilizers in solution.^[33,34] In our case, one hemisphere consists of PS chains (miscible with PPE), the other one contains PMMA chains (miscible with SAN) and the cross-linked PB core (immiscible with PS/PPE and PMMA/SAN) is located at the PPE/SAN interface. Bahrami et al.^[21] investigated the superior effect of JPs on the morphological development of PPE/SAN immiscible blends in comparison with the SBM triblock terpolymers processed by extrusion.

The aim of this work is therefore to analyze the effect of Janus particles compatibilization on the foam morphology and density of PPE/SAN polymer blend foams. Since these particles provide finer morphologies compared to other compatibilizer alternatives, and have higher interfacial activities, it is interesting to investigate their effect during a complex foaming process. Especially, the foaming behavior between the blend partners is investigated by transmission electron microscopy (TEM) and scanning electron microscopy (SEM) to quantify the effect on the morphology.

2. Experimental Section

2.1. Materials

Poly(2,6-dimethyl-1,4-phenylene ether) (PPE PX100F) was provided by Mitsubishi Engineering Plastic Corporation as a powder with a M_w of 12.9 kg mol^{-1} and a dispersity of $D = 1.63$. Poly(styrene-*co*-acrylonitrile) SAN (Luran VLL19 100), with an AN content of 19 wt% was provided by BASF AG. The SAN has an M_w of 122 kg mol^{-1} and a dispersity of

$D = 2.13$, which was measured via size exclusion chromatography (SEC) using THF as solvent and narrowly distributed PS standards.

Janus particles (JP), used as compatibilizer with 1, 2, 5, and 10 wt% in PPE/SAN blends (60/40 (w/w) blend ratio), were synthesized from $S_{40}B_{31}M_{29}^{108}$ triblock terpolymers (M_w of the SBM triblock is 108 kg mol^{-1} , and subscript represents content of each block in wt%) according to a previously published method.^[21,32]

In order to prevent any degradation during processing, 0.1 wt% stabilizers were added to the blend component mixtures before the extrusion process. The stabilizer formulation consisted of 67 wt% Irganox 1010 and 33 wt% Irgafos 168 provided by BASF SE.

2.2. Manufacturing Methods

PPE and SAN were dried in a vacuum oven for 24 h at 80°C , and in order to avoid further cross-linking of the remaining not cross-linked double bonds in the PB core of the JPs, the latter were dried at 40°C in vacuum. The blend systems were compounded with a Brabender Lab Station, which is a co-rotating twin-screw extruder with a screw diameter of 20 mm and a screw length of 600 mm ($L/D = 30$). During compounding, the average melt temperature and the screw speed were set at 250°C at 85 rpm, respectively, resulting in a throughput of 1 kg h^{-1} using a gravimetric feeding. After extrusion, the blends were air-cooled and pelletized.

The PPE/SAN blends were compression molded at 260°C (pure PPE at 270°C and pure SAN at 160°C) into samples of $18 \times 10 \times 1 \text{ mm}$ in size, using a hot hydraulic press (Paul Weber). The materials were pre-heated for 5 min without pressure, then exposed to 100 kN force for 5 min and cooled for 2 min under 20 kN force in a cold press.

2.3. Morphological Characterization

A transmission electron microscope, TEM Zeiss 922 Omega, was used to determine the morphology of the blends after extrusion. The ultrathin cuts were prepared with a Reichert-Jung ultra-microtome (Ultracut E) and then stained with osmium tetroxide (OsO_4) for 2 h at ambient pressure. After staining, SAN appears as the light phase, and PPE appears as the dark phase. Due to staining of residual, non-cross-linked double bonds in the PB core of the JPs with OsO_4 , the JPs are visible as black dots.

2.4. Differential Scanning Calorimetry

The glass transition temperature (T_g) of the neat materials and the immiscible blend systems were measured using a TA Instrument DSC Q1000. The method consists of a heating-cooling-heating cycle under nitrogen atmosphere from 25 to 250°C at a scanning rate of 10 K min^{-1} . The values of the second heating cycle were evaluated to calculate the T_g .

Additionally, modulated DSC measurements at low temperature were done with pure JPs and with PPE/

SAN + 10 wt% JP to analyze the influence of compatibilization and to determine the effect of cross-linking of the PB part of JP. The measured temperature range was -150 to 150°C at a heating rate of $3 \text{ K min}^{-1} \pm 1 \text{ K}$ under nitrogen atmosphere. This method gives information of the reversing and non-reversing characteristics of thermal events. Especially, the polybutadiene part in the blend with low amount can be detected.

2.5. High-Pressure DSC (HP-DSC)

In order to understand the plasticizing effect of CO_2 under pressure on the glass transition temperature, the specimens were analyzed with a high-pressure DSC 204 HP Phoenix from Netzsch GmbH. The samples were saturated with CO_2 at given pressures of (1, 10, 35 bar) for at least 24 h. The measurement was performed from 25 to 260°C with a heating rate of 10 K min^{-1} under nitrogen atmosphere. The first heating cycle was used for evaluation. Due to technical difficulties, it was not possible to measure SAN at 35 bar (because of its lower T_g) and the other materials at 40 bar, which is the CO_2 mass uptake pressure.

2.6. Mass Uptake and Solubility Measurements

The solubility and diffusion coefficients of the blowing agent CO_2 were measured in mass uptake experiments according to the method of Berens et al.^[35] Therefore, samples ($18 \times 10 \times 1 \text{ mm}^3$) were placed in an autoclave under CO_2 at 40 bar while keeping the temperature constant at 40°C . Those parameters have been chosen due to the lower critical solution temperature (LCST) behavior of PMMA, which partially exists in JPs and therefore at the blend interface.^[36] The mass uptake was then measured in several time intervals, and the increase in the sample weight indicated the amount of CO_2 dissolution in the polymer.

The mass uptake was calculated by dividing the total amount of gas that diffused into the polymer by the sample weight. The ratio of mass at the time t and the mass at equilibrium (M_t/M_{eq}) was calculated with following Equation 1 of Crank:^[37,38]

$$\frac{M_t}{M_{\text{eq}}} = \frac{2}{\sqrt{\pi}} \frac{D^{1/2}}{l} \sqrt{t} \quad (1)$$

M_t : Mass of absorbed blowing agent at time t in mg

M_{eq} : Mass of the sample at equilibrium in mg

D : Diffusion coefficient in $\text{cm}^2 \text{ s}^{-1}$

l : Sample thickness in cm

t : Time in s

Then, M_t/M_{eq} was plotted over the square root of desorption time in order to determine the slope of this curve (S). By determining the slope of the curve the diffusion coefficient could be calculated with following Equation 2:^[38]

$$D = \frac{\pi}{4} \cdot S^2 \cdot l^2 \quad (2)$$

Table 1. Parameters for batch foaming experiments.

Foaming time	Foaming temperature			
10 s	120 °C	140 °C	155 °C	170 °C
30 s	120 °C	140 °C	155 °C	170 °C

2.7. Foaming Experiments

2.7.1. Temperature-Induced Batch Foaming

Based on the HP-DSC results, four temperatures between the T_g of PPE and SAN were chosen as foaming temperatures (Table 1). Blends were foamed via a temperature-induced foaming method; firstly, the specimen are placed in a high-pressure vessel (at 40 bar and 40 °C) and loaded with carbon dioxide, as blowing agent. In the saturation step, the samples were kept under pressure for 24 h to ensure the equilibrium gas absorption; afterwards, the pressure in the vessel was rapidly released to atmosphere. Immediately then, they were put into a silicon oil bath at the relevant foaming temperature. The foam structure was stabilized, cell growing as well as cell coalescence was prevented, by cooling to room temperature with another silicon oil bath. The foaming time (sample exposure in the hot oil bath) was varied between 10 and 30 s in order to optimize the processing parameters based on the foam densities.

2.7.2. Characterization of the Cellular Structure

The morphology of the cellular structure was analyzed using normal and field emission scanning electron microscopes, SEM (Jeol JSM-IC 848) and FESEM (Zeiss Leo 1530 Gemini), at an acceleration voltage of 15 and 3 kV, respectively. The samples were sputtered with a 1.3 nm layer of platinum prior to imaging. The average cell diameter and cell size distribution of the polymer blend foams were measured after analyzing the micrographs with ImageJ software.

The foam density was measured by the buoyancy method according to ISO 1183 using the AG245 from Mettler Toledo. For a representative value, at least three samples from each material were analyzed and the average was reported.

3. Results and Discussion

3.1. Effect of JP Compatibilization on the Blend Morphology

The morphology of the neat and compatibilized blends with 1, 2, 5, and 10 wt% Janus particles is extensively discussed in ref.^[21] It was shown that the Janus particles are located at the interface and by reducing the interfacial tension between the immiscible PPE and SAN polymers, further reduced the PPE domain size in the SAN matrix (see Figure 1).

Since the Janus particles are used as foam nucleating agents in the current study, Figure 1 shows their location at the interface at higher magnifications. It was found that the less

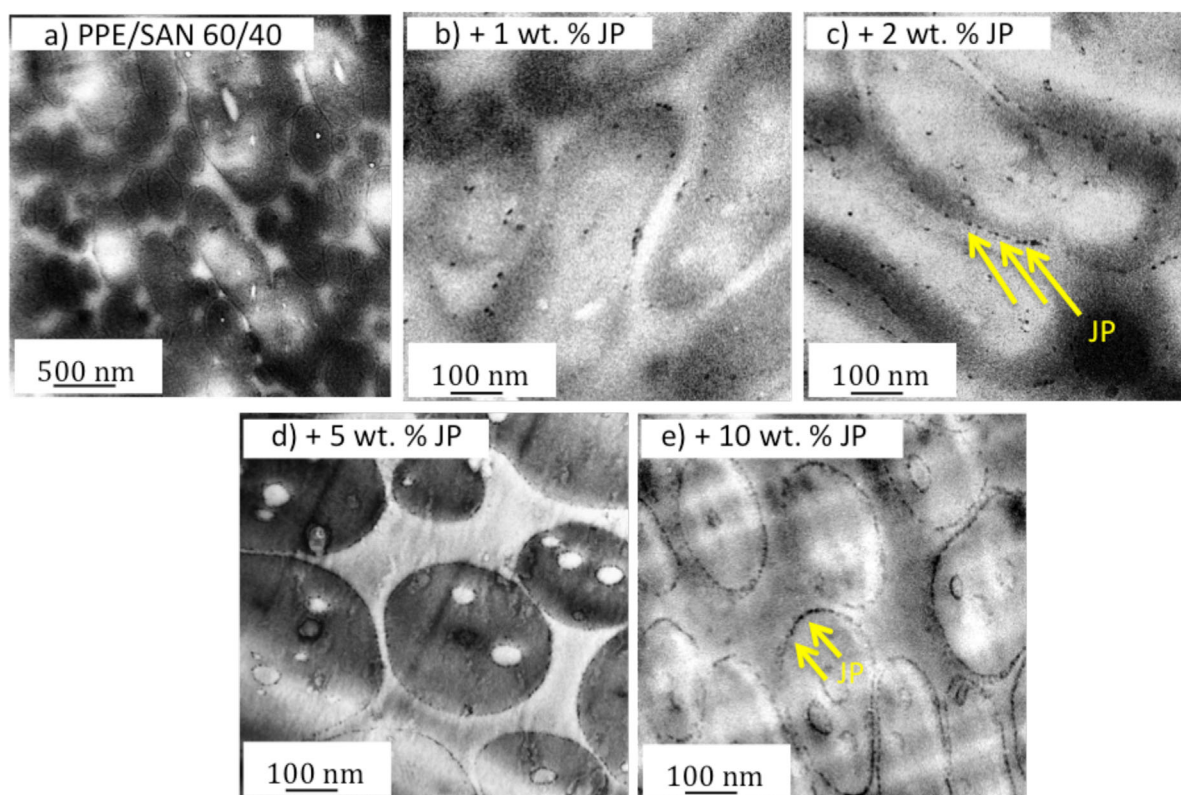


Fig. 1. TEM micrographs of Janus particles location at the interface of compatibilized PPE/SAN 60/40 blends: (a) irregular neat blend, (b) and (c) homogenized blends with large droplets containing 1 and 2 wt% JPs, respectively, (d) and (e) small, homogenized PPE droplets.

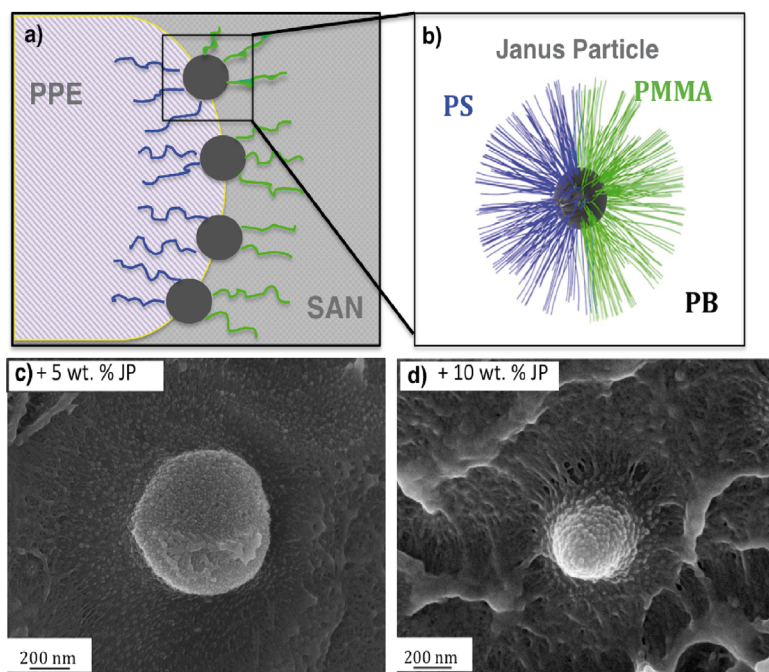


Fig. 2. (a) and (b): schematic model of the raspberry morphology, JPs are located at the interface of PPE and SAN; (c) and (d): FESEM micrographs of the fractured surface of PPE/SAN 60/40 (w/w) blends with a PPE droplet covered by JPs (little bright dots on the PPE droplet).

viscous phase (SAN) tends to form a continuous matrix and the more viscous phase (PPE) occurs as dispersed domains.^[39] Due to staining with OsO_4 , the PPE phase appears as the gray phase, and unstained SAN is the lighter phase. The residual, not cross-linked butadiene parts of the JP cores are also stained and are visible as black dots. The difference in the PPE color contrast is due to differences in TEM sample thickness. The TEM pictures show that Janus particles are exclusively located at the interface of the PPE/SAN blends. While the neat blend without compatibilization shows irregular-shaped domains of PPE and SAN (Figure 1a), adding only 1 wt% JPs (Figure 1b) results in more homogeneous, yet relatively large PPE domains with few JPs at the interface. Addition of 2 wt% JPs (Figure 1c) shows visibly more JPs at the interface. This leads to some reduction of the PPE droplet diameter and a more homogeneous distribution of the droplets, which indicates increased compatibility of the blend system.^[21] By further increasing the JP amounts (Figure 1d and e), from 5 wt % up to 10 wt%, their numbers is enough to fully cover the PPE droplets and reduce their sizes significantly. This is important, since the structure with the PPE phase fully covered with JPs, called raspberry morphology, can later on hinder gas entrance in the PPE phase during the saturation step before foaming. This may happen due to the fact that the cross-linked PB core of the JPs has a significantly lower CO_2 solubility compared to the other materials and can act as a barrier layer. Figure 2 shows the schematic structure of the raspberry morphology in the blends with 5 and 10 wt% JPs. The structure of the Janus particles (PB core and the PS and PMMA side) is also shown in a zoom-in image. The presence

of SAN micelles in the PPE phase has been extensively discussed in ref.^[21]

Adding 10 wt% JP (Figure 1e), the blend morphology reaches the ultimate droplet homogeneity and the PPE droplet diameter is further decreased to around 300 nm (± 160). The reduction of the blend domain size by compatibilization was also investigated in the following publications.^[17,18,20]

There are also few JPs available in the PPE matrix in Figure 1, especially in blends with 1 and 2 wt% JP, this can be explained by the higher affinity of the PPE/PS miscible phases compared to the SAN/PMMA as explained in detail previously.^[21] The PB middle block (black dots) is located at the interface of the blend because it is immiscible with all polymers and covalently connected to PS and PMMA and therefore produces the known raspberry structure shown in Figure 2a and b. In picture c and d of Figure 2, SEM images of the blends compatibilized with 5 and 10 wt% JPs are shown. A single PPE domain, which is fully covered by JPs, is shown for each blend. Firstly, the difference in the droplet size depending on the amount of JPs is visible, as the PPE droplet in the blend with 5 wt% JPs is larger than that with 10 wt %. In addition, the single JPs are visible on the surface of the PPE droplets appearing as white spots.

3.2. Effect of Compatibilization on Thermal Properties

The glass transition temperature plays an important role during temperature-induced foaming. The T_g values of the neat and compatibilized blends are shown in Table 2. For each blend, two T_g values correlating to the PPE and SAN phase are determined. It is expected that after compatibilization, the T_g values change, reducing for PPE, since it is miscible with PS that has a significantly lower T_g of around 100 °C. In case of the SAN phase, due to their local miscibility with PMMA that has almost the same value of T_g (around 110 °C), no significant change is expected. The higher the amount of JPs in the blend system, the lower the glass transition temperature of PPE in the compatibilized-blend systems. This is due to the higher

Table 2. Thermal properties of neat PPE and SAN, as well as the corresponding blends.

Material	T_g in °C of SAN	T_g in °C of PPE
SAN	110 $\pm$ 0.4	–
PPE	–	215 $\pm$ 0.5
PPE/SAN neat	109 $\pm$ 0.2	210 $\pm$ 0.5
+1 wt% JP	109 $\pm$ 0.5	209 $\pm$ 0.1
+2 wt% JP	109 $\pm$ 0.5	208 $\pm$ 1.3
+5 wt% JP	110 $\pm$ 0.6	206 $\pm$ 0.8
+10 wt% JP	111 $\pm$ 0.8	204 $\pm$ 1.3

amounts of PS (with lower T_g) that is available in the system as previously also reported for other systems.^[40]

Additionally, the T_g of the JP core (partially cross-linked PB) is investigated, since it is located directly at the interface and acts as the nucleating site during foaming. While synthesizing JPs from SBM triblock terpolymers, the PB core is cross-linked using a UV cross-linker.^[32] As there is only 31 wt% of PB in the JPs, corresponding to only ca. 3 wt% in the PPE/SAN blend with 10 wt% JPs, the T_g of the JP's core was measured using a more sensitive method (modulated DSC) for the neat JPs and the JP-compatibilized blends. The non-cross-linked PB block in the SBM precursor has a lower T_g (-72°C), while the cross-linked PB core of the JPs has a higher T_g of around 0°C . Nearly the same values for PB are detected by analyzing the PPE/SAN blend with 10 wt% JP. The high standard deviation of around 6°C can be due to the distribution of JP in the polymer blend and variations of their cross-link degrees.

3.3. Effect of CO_2 Solubility and Diffusion Coefficient

The first step during the foaming process is the absorption and monomolecular dissolution of the blowing agent into the polymer. Hence, mass uptake measurements of CO_2 in the neat and compatibilized blends were performed. The CO_2 mass uptake of the neat materials compared to the neat PPE/SAN blend is shown in Figure 3. PPE has a higher CO_2 mass uptake than SAN. As expected by the rule-of-mixture, the CO_2 mass uptake of the 60/40 polymer blend is located in between the neat materials at around 7.3 wt%. The CO_2 uptake of the blend is closer to the SAN value due to the fact that SAN is the continuous phase in the blend and dominates its behavior.

By adding JPs, the carbon dioxide uptake is at first increased, however, careful analysis of Figure 3 shows that the blends with 5 and 10 wt% JPs do not follow this trend. This can be explained by looking more deeply at the morphologies of the compatibilized blends. When the PPE domains are partially covered by JPs (in case of 1 and 2 wt% JPs), the

CO_2 presence at the PPE/SAN interphase can produce internal pressure. As the stress concentration increases at these points, this pressure can produce microcrazes via cleavage fracture (and in some cases even cracks if microcrack propagation cannot be stopped via smaller PPE domains) in the material. As soon as these crazes/cracks form in the sample, more surface is produced, hence, the surface available for the CO_2 absorption is increased, and higher amounts of CO_2 will be absorbed in the material. Consequently, larger mass uptake values are observed for blends with 1 and 2 wt% JPs. In case of blend with 5 and 10 wt% JPs, since the phases are fully connected via compatibilizer (due to full coverage of the PPE phase with JPs), there are no weak stress concentration points and as a result no craze/cracks to provide more surface in the material. In addition, the partially cross-linked PB has a lower CO_2 solubility compared to the other phases.^[41] In case of blends with fully covered PPE domains, it could act as a barrier layer and hinder the CO_2 diffusion into the PPE phase. A schematic illustration of the assumption, alongside an optical microscope image of the sample with 1 wt% JPs showing microcracks after loading with CO_2 is shown in Figure 4. In addition, if PPE, which has a higher CO_2 solubility, is fully covered with JPs, then the ability of CO_2 diffusion is reduced since it has to go through an extra layer of compatibilizer (here containing a partially cross-linked PB phase).

The diffusion coefficient determines how fast the blowing agent diffuses into the polymer. Considering the apparent diffusion coefficients of PPE and SAN and the effect of blending, solubility of CO_2 in materials was calculated and can be seen in Table 3.

In the literature, it is known that carbon dioxide has a plasticizing effect on amorphous polymers, resulting in a reduction of the glass transition temperature.^[42–45,47] The extent of T_g decrease depends on the pressure and the concentration of the dissolved blowing agent, the type of blowing agent used and its affinity to the polymer. The

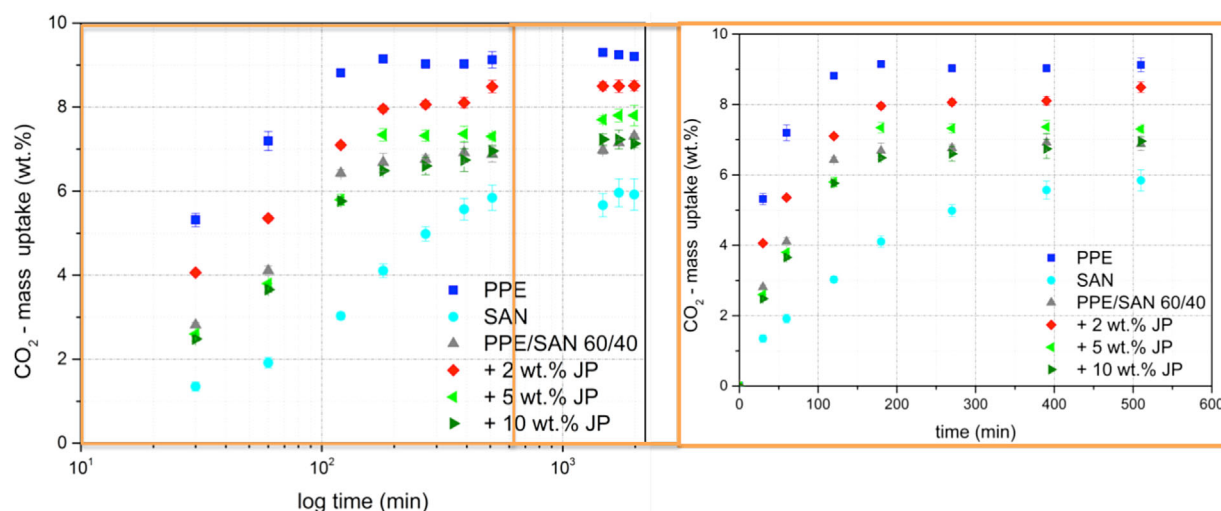


Fig. 3. CO_2 mass uptake in dependence of time for PPE, SAN, and PPE/SAN 60/40 (w/w) blends compatibilized by JPs.

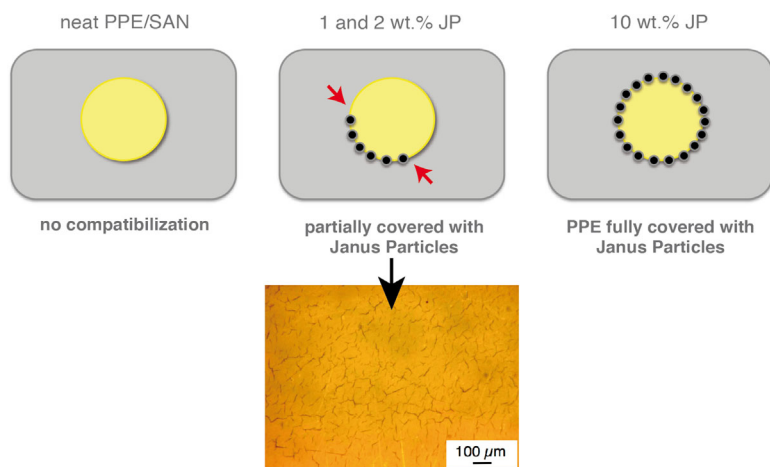


Fig. 4. Effect of compatibilizer on the CO₂ sorption behavior of PPE/SAN blends, stress concentration at areas marked by red arrows leads to microcrazes in CO₂-loaded blends with, e.g., 1 wt% JP.

Table 3. Solubility and apparent diffusion coefficients of CO₂ in neat and blend systems of PPE and SAN.

Material	Solubility in [mg g ⁻¹]	Diffusion coefficient in [cm ² s ⁻¹]
PPE	94 ± 3.5	1.20 × 10 ⁻⁶ ± 3.6 × 10 ⁻⁸
SAN	62 ± 5.8	0.29 × 10 ⁻⁶ ± 1.4 × 10 ⁻⁸
PPE/SAN	73 ± 6.4	0.77 × 10 ⁻⁶ ± 7.4 × 10 ⁻⁸
neat		
+1 wt% JP	87 ± 4.4	1.20 × 10 ⁻⁶ ± 2.0 × 10 ⁻⁹
+2 wt% JP	85 ± 4.2	1.10 × 10 ⁻⁶ ± 6.8 × 10 ⁻⁹
+5 wt% JP	76 ± 2.3	0.23 × 10 ⁻⁶ ± 0.4 × 10 ⁻⁹
+10 wt% JP	71 ± 4.9	0.61 × 10 ⁻⁶ ± 5.0 × 10 ⁻⁹

absorbed gas increases the free volume and therefore enhances the mobility of the chain segments.^[49] Therefore, the T_g of the neat polymers as well as the neat blend was measured under CO₂ atmosphere at different pressures. Figure 5 shows the dependence of the glass transition temperature on CO₂ pressure. The glass transition temperature decreases with CO₂ pressure, due to its plasticizing effect.

The glass transition temperature of neat PPE is reduced by 24 °C, at a CO₂ pressure of 40 bar (extrapolated values). In case of neat SAN, large extrapolations should be avoided since more points in between are missing, but a similar trend is expected. The glass transition temperature can be correlated to the CO₂ solubility in the material, i.e., a higher CO₂ solubility results in a more pronounced decrease in T_g .^[48] This effect, with a similar trend, is also visible for the polymer blend. O'Neil and Handa^[46] and Zhang and Handa^[49] investigated the T_g depression of PS by using carbon dioxide. The glass transition temperature of PS decreased around 30 °C at a pressure of 40 bar. The T_g of PET decreased about 50% at a CO₂ pressure of 30 bar.^[50] A comparison to the results in this work concerning the plasticizing effect of CO₂ on PPE can be found in a publication of Handa et al.^[45] He measured a glass

transition temperature for PPE (provided by General electric) of $T_g = 195.8$ °C at 40.8 bar CO₂, and the slight deviation to this work could be due to the difference in the molecular weight of the PPE used in both works. This information will help us to define the optimum temperature range for foaming of the blends. The T_g of the PPE phase is 180 °C during saturation and later foaming, hence, below this temperature blends have enough stability due to the still rigid PPE domains.

3.4. Optimizing the Foaming Process

At first, the effect of foaming time was investigated, defining the density of the foam as the quantifying factor to compare the foamed materials. Therefore, the foaming time at which the lowest foam density was achieved was defined as the optimum. Hence, the PPE/SAN blend systems, with and without compatibilizer, were foamed for 10 and 30 s at the previously defined temperatures (Table 1, Experimental Section). The lower foam densities were correlated to the shorter foaming time of 10 s, therefore it was chosen as the optimum value. It is expected that the longer foaming time of 30 s would lead into further cell growth and eventually cell coalescences, resulting in thicker cell walls, collapsed cells and, as a result, possibly higher densities.^[49,50]

The influence of foaming temperature on the density of the neat materials and the blends is shown in Figure 6 and 7a. The highest density reduction of the SAN (89%) is reached at a foaming temperature of 140 °C. Although the employed foaming temperatures are below the glass transition temperature of PPE (even after plasticization), the neat PPE still gets slightly foamed (with very small cell sizes) at 140 and 155 °C, which resulted in a density reduction of about 20%. As expected, the foam density of the PPE/SAN blend is located in between the results of PPE and SAN (Figure 6). The neat

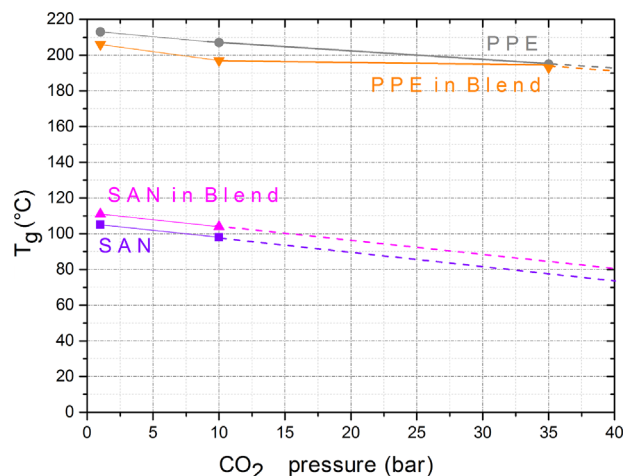


Fig. 5. Plasticizing effect of carbon dioxide on PPE, SAN, and PPE/SAN 60/40 (w/w) blend system.

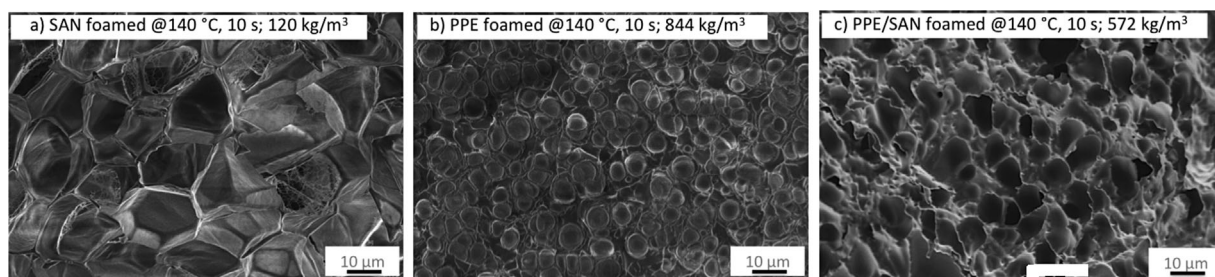


Fig. 6. Foam morphology of (a) SAN, (b) PPE, and (c) PPE/SAN, foamed at 140 °C for 10 s.

polymers as well the blend do not show any density reductions at 170 °C (hence not shown in the graph) probably due to the low viscosity of the SAN phases at this temperature, giving rise to coalescence of the foam cells during foaming.

In Figure 7b, the effect of compatibilization on the foam density of the blend at the defined foaming temperatures is illustrated for the neat PPE/SAN blend and with 1 and 10 wt% JPs as representatives of both morphologies (PPE droplets partially and fully covered with JPs, respectively).

The blend with 10 wt% JPs shows higher densities compared to the neat blend and the blend with 1 wt% JPs. Because of the higher JP content the nucleation efficiency is improved, which reduces the size and therefore increases the surface of the PPE domains, which can also act as nucleating sites. This probably results in smaller cell sizes (due to nucleation) and eventually higher densities (due to smaller CO₂ solubility and diffusion coming from the raspberry morphology). The density reduction in case of the blend with 1 wt% JP can be correlated to the strong interface provided by the Janus particles, which leads to a homogeneous growth of cells. Looking at the effect of foaming temperature, the highest density reduction was achieved at 140 °C. The foaming temperature of 155 °C increases the densities again due to lowering the viscosity of the blend materials causing

destabilization of the cells, cell collapse, and thicker cell walls. In order to correlate the effect of compatibilization on the foaming process in more detail, a foaming temperature of 140 °C, which produces foams with the lowest densities, is chosen and corresponding SEM images of the foams are shown in Figure 8.

It is important to note that without added Janus particles, the PPE/SAN blend can barely foam homogeneously at 140, 155, or 170 °C (see Figure 6, 8a). Foaming of the SAN phase brings several holes in the structures, which result in density reduction. However, this is not a defined cellular structure one would expect from an efficient foaming process. Adding JPs initiates cell nucleation and, first, a low JP content of 1 wt% in PPE/SAN leads to larger cells from the SAN parts, as well as very small cells related to the PPE phase that is not able to foam entirely. The PPE phases are not fully attached to the SAN matrix and therefore cannot expand with them into larger cells. Hence, there are unfoamed areas correlating to the PPE phase visible in the SEM pictures (Figure 8b).

Comparing to the neat blend (Figure 8a), one can see that the foam structure is improved, but not yet homogenous. With 2 wt% JPs, the average cell size stays the same at 1.8 µm (±0.8 µm); however, the number of unfoamed PPE domains is reduced (Figure 8c). Figure 9 shows the cell size distributions

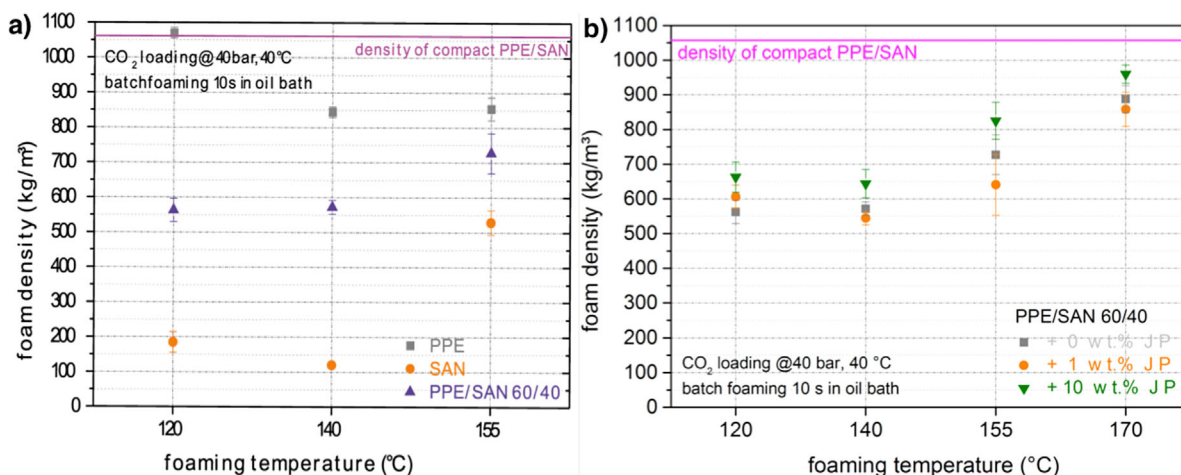


Fig. 7. Effect of foaming temperature on the foam density for (a) neat materials and (b) with JP-compatibilized PPE/SAN blends.

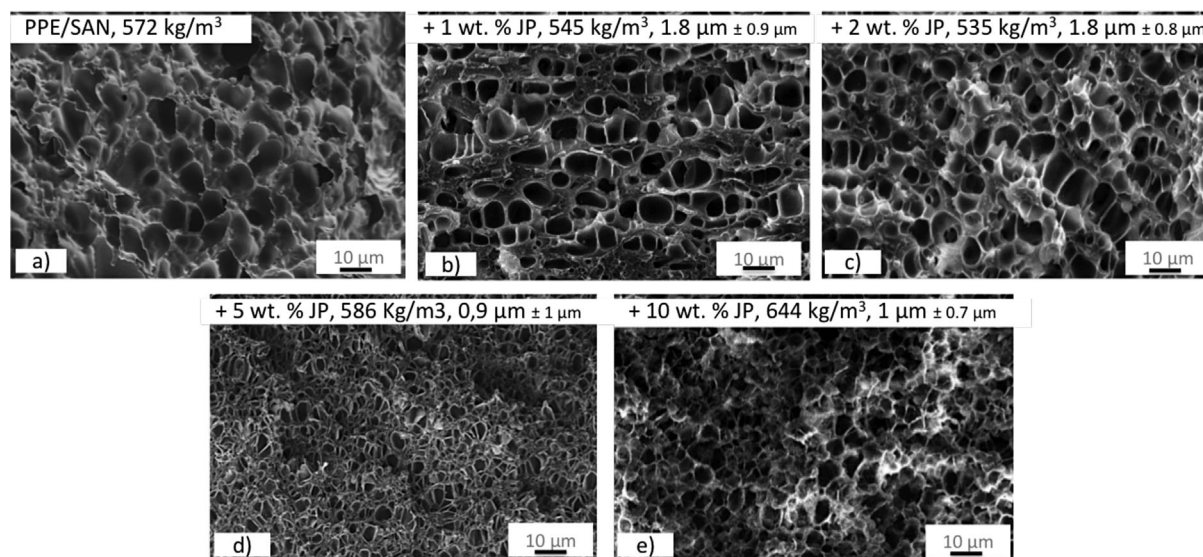


Fig. 8. SEM micrographs showing the effect of JP compatibilization on the morphology of PPE/SAN (60/40 w/w) blend foams at different JP contents (foaming at 140 °C for 10 s).

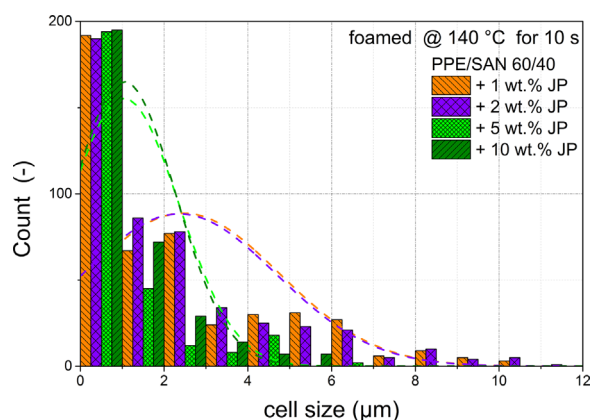


Fig. 9. Cell size distributions of compatibilized PPE/SAN (60/40 w/w) blend foams at different JP contents (foaming at 140 °C for 10 s).

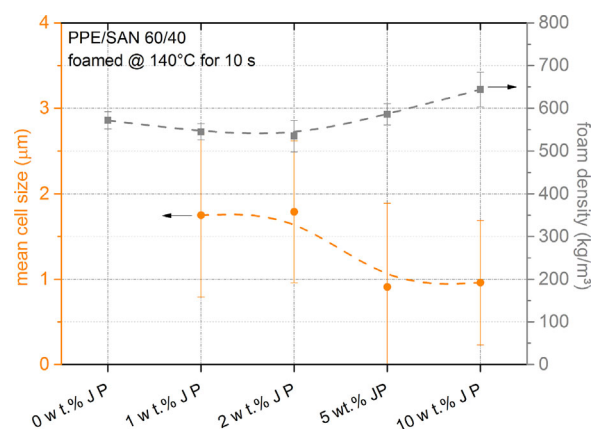


Fig. 10. Correlation of mean cell size and foam density with a focus on the effect of compatibilization of PPE/SAN blend with JP.

of the compatibilized blends, one can see that the blends with 1 and 2 wt% JPs show a similar, relatively wide distribution compared to the blends with higher compatibilizer contents.

Polymer foams are mainly characterized via their mean cell size and foam density. Therefore, the correlations of the mean cell size and the foam density for the optimized PPE/SAN blend foams compatibilized with JPs at 140 °C and 10 s are shown in Figure 10. The foam density is slightly decreasing with increasing JP content from 0 to 2 wt% due to the increase in CO₂ solubility by the mechanism explained earlier. The high amount of JPs at 5 and 10 wt% is reducing the free volume of the system and therefore reduces the solubility of the material. Thus, from the point that full Raspberry structures are formed in the morphology, the foam density of the blend increases. On the contrary, the mean cell size is visibly decreasing in the blend with 5 and 10 wt% JPs compared to the blend with lower amounts of JPs. The increase in compatibilizer content reduces the PPE domain

size, which also increases the surface available between the blends for the nucleation. The higher content of JPs also increases the interfacial interaction between PPE and SAN, which leads to additional foaming initiated at the interface, generating smaller cells.

At higher JP contents than 2 wt%, the foaming capability of the PPE phase is improved since it is mechanically stronger coupled to the SAN matrix (larger number of JPs at the interface, Figure 1). As already mentioned before, the neat foam blend nearly did not foam, therefore it was not possible to evaluate the morphology properly. With 5 and 10 wt% JP, the foam homogeneity is further improved and also the average cell diameter is reduced to 1 μm and even below 1 μm with 5 wt% JP (Figure 8c, d, and 10). This can be attributed to the strong mechanical coupling between the PPE and SAN phases, provided by the full coverage of the interface by JPs, and to the high number of JPs acting as nucleation sites. Gong and Ohshima^[16] discussed the effect of interfacial tension on

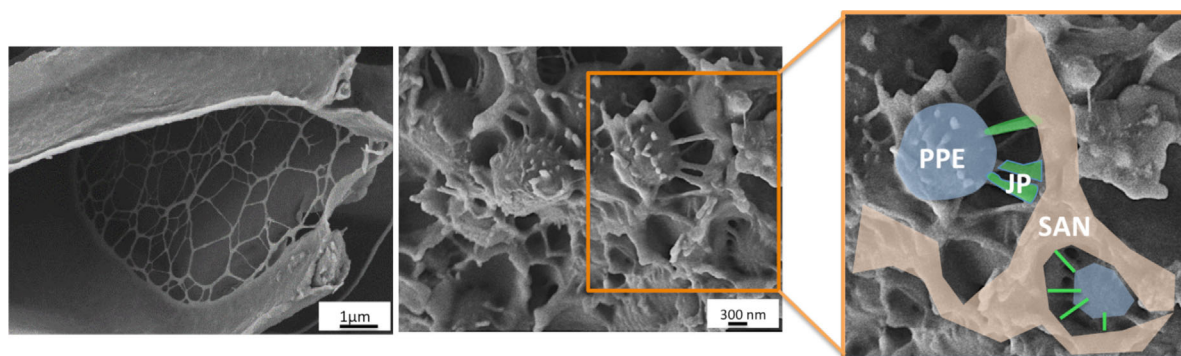


Fig. 11. FESEM micrographs of PPE/SAN compatibilized with 10 wt% JP showing open cells and spider webs (foamed at 140 °C for 10 s), in the right picture a small PPE domain (round particle) is attached to the SAN matrix by Janus particles.

the bubble nucleation. They discovered that a low interfacial tension, which is here achieved through JP compatibilization, required more energy for nucleation. Therefore, the cell size at the interface is in general much smaller than in the SAN matrix.

In the FESEM micrographs of compatibilized PPE/SAN blends (60/40 w/w), open cell structures (spider web-like) are visible in the SAN matrix and at the blend interface. This fine cellular structure of compatibilized PPE/SAN blends is illustrated in Figure 11.

The open cell structure is developed through matrix stretching due to expansion and stress concentration at the interface. The Janus particles are located at the interface and the PPE domains can act as nucleation sites, whereby the Janus particles are strongly connecting the PPE domains to the SAN matrix. The SAN phase plus the JPs attached to it stretch and form spider-like webs due to the strong interface and therefore increase the open cell content. The CO₂ dissolved between the SAN and PPE phase starts growing a bubble and stretches the Janus particles (plus the softer SAN phase attached to it), resulting in fiber-like structures. The control of nucleation and open cell content with a second blend phase, which results in finer cell structures has been investigated in the literature.^[7,51–54] In the current study, the cell opening phenomena can occur by the use of JPs as very efficient nucleation agents, or by the combination of two blend components with different glass transition temperature, whereby the hard PPE phase (higher T_g) is dispersed in the soft (lower T_g) SAN matrix. During expansion, the matrix is stretched and the tougher phase retains its shape. When both phases have a poor bonding and different deformation behaviors, stress concentration at the interface may cause cell wall rupture between the hard and the soft phase. Here, the JPs are keeping the PPE and SAN phases connected. The same mechanism explained for another class of polymers was extensively discussed by Park et al.^[55] In LDPE/PS blends, one component (LDPE, low-density polyethylene) is semi-crystalline and during the foam extrusion process, the hard crystalline part plays a similar role as the hard PPE phase in the compatibilized PPE/SAN system.

4. Conclusions

Immiscible PPE/SAN blends (60/40) were successfully compatibilized by Janus particles, which led to the formation of a raspberry morphology in the compact blend. The reduction in the PPE domain diameter after compatibilization was significant and lead to very fine dispersions of the second phase up to 300 nm in diameter, which gives them the potential to be used as nucleating sites during foaming.

The sorption measurements showed that the carbon dioxide solubility in the solid blend at RT increased with adding up to 2 wt% JPs due to extra surface appearing by crazes at the interface. Due to the raspberry morphology achieved with 5 and 10 wt%, the stress concentration points where these cracks initiated did not exist and therefore the CO₂ solubility decreased. The diffusion coefficient showed the same trend and was reduced for higher compatibilizer contents (starting with 5 wt% JP) due to the barrier effect of the partially cross-linked PB core of the JPs. The optimum foaming conditions (time and temperature) for the compatibilized blends were established, and the effect of compatibilization on the foam morphology, average cell size, and the density of the foams was investigated. The average foam cell size decreases with increasing amount of JP, due to the fact that smaller PPE domain sizes provide increased foam nucleation sites for foaming. JPs were also initiating cell nucleation in blend foams and due to their strong attachment between the phases during foaming, fibril-like structures were produced resulting in foams with partial open cell contents. The lowest cell size (900 nm) with a homogeneous cell structure is achieved with the blend compatibilized with 5 wt % JPs. The addition of more JPs (10 wt%) lead to densely packed PPE domains with very small inter-domain distances hindering further foam growth. The minimum foam density was achieved with the blend containing 2 wt% JPs (535 kg m⁻³), showing cell diameters of 1.7 μm. In this case, the SAN phase is dominantly foamed into larger cell sizes due to weak attachment of the PPE phase to it. In the end, it was concluded that a foaming temperature of 140 °C and foaming time of 10 s is the best combination to obtain homogeneous

foams of PPE/SAN blends compatibilized with JPs, with very small cell sizes of 1.8–0.9 μm but at relatively high densities of 350–650 kg m^{-3} . One can conclude that using JPs with high interfacial activities would promote better nucleation compared to the neat blend during foaming. At the same time, the strong bonding of the JPs due to their high activity would allow for production of open cellular foam structures.

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